

Metric Spaces: Complete Lecture Notes

(Definitions, Examples, Theorems, and Detailed Proofs)

Contents

1 Preliminaries: Sets and Functions	2
2 Lecture 1: Metrics and Metric Spaces	2
2.1 Examples of metrics	2
2.2 Basic inequalities in metric spaces	3
2.3 Constructing new metrics	3
3 Lecture 2: Standard Metric Spaces	5
3.1 Metrics on $\mathbb{R}^n$ and comparison of norms	5
3.2 Sequence spaces ℓ_p	5
3.2.1 Hölder and Minkowski (with proofs)	6
4 Lecture 3: Topology Induced by a Metric	7
4.1 Open sets, closed sets	7
4.2 Closure, interior, boundary	8
5 Lecture 4: Density and Examples in $\mathbb{R}$ and $\mathbb{R}^2$	9
6 Lecture 5: Sequences, Limits, and Cauchy Sequences	10
6.1 Definitions	10
7 Lecture 6: Completeness	11
7.1 Completeness of ℓ_p (full proof)	11
7.2 Closed subsets and completeness	12
8 Lecture 7: Continuity and Uniform Continuity	12
8.1 Continuity	12
8.2 Uniform continuity	13
9 Lecture 8: Compactness, Total Boundedness, Sequential Compactness	14
9.1 Compactness	14
9.2 Sequential compactness $\Leftrightarrow$ completeness + total boundedness	15
10 Lecture 9: Separability	16
11 Lecture 10: Banach Fixed Point Theorem	16
12 Lecture 11: Cantor Intersection and Perfect Sets (optional enrichment)	17
13 Appendix: A Quick Checklist of Core Results	18

1 Preliminaries: Sets and Functions

Definition 1.1 (Function, image, preimage). Let X, Y be sets. A *function* $f : X \rightarrow Y$ assigns to each $x \in X$ a unique element $f(x) \in Y$. For $A \subset X$, the *image* of A is

$$f(A) := \{f(x) : x \in A\} \subset Y.$$

For $B \subset Y$, the *preimage* of B is

$$f^{-1}(B) := \{x \in X : f(x) \in B\} \subset X.$$

Lemma 1.2 (Preimage rules). Let $f : X \rightarrow Y$ and $\{B_\alpha\}_{\alpha \in I} \subset \mathcal{P}(Y)$. Then:

$$f^{-1}\left(\bigcup_{\alpha \in I} B_\alpha\right) = \bigcup_{\alpha \in I} f^{-1}(B_\alpha), \quad f^{-1}\left(\bigcap_{\alpha \in I} B_\alpha\right) = \bigcap_{\alpha \in I} f^{-1}(B_\alpha),$$

and for any $B \subset Y$,

$$f^{-1}(Y \setminus B) = X \setminus f^{-1}(B).$$

Proof. We prove the union identity; the others are similar.

Let $x \in f^{-1}(\cup_\alpha B_\alpha)$. Then $f(x) \in \cup_\alpha B_\alpha$, so $f(x) \in B_{\alpha_0}$ for some α_0 . Thus $x \in f^{-1}(B_{\alpha_0}) \subset \cup_\alpha f^{-1}(B_\alpha)$.

Conversely, if $x \in \cup_\alpha f^{-1}(B_\alpha)$, then $x \in f^{-1}(B_{\alpha_0})$ for some α_0 , i.e. $f(x) \in B_{\alpha_0} \subset \cup_\alpha B_\alpha$. Hence $x \in f^{-1}(\cup_\alpha B_\alpha)$. $\square$

2 Lecture 1: Metrics and Metric Spaces

Definition 2.1 (Metric and metric space). A *metric* on a nonempty set X is a function $d : X \times X \rightarrow [0, \infty)$ such that for all $x, y, z \in X$:

(M1) $d(x, y) = 0 \iff x = y$ (identity of indiscernibles),

(M2) $d(x, y) = d(y, x)$ (symmetry),

(M3) $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

Then (X, d) is called a *metric space*.

Remark 2.2. From the axioms, we always have $d(x, y) \geq 0$ and $d(x, x) = 0$.

2.1 Examples of metrics

Example 2.3 (Usual metric on $\mathbb{R}$). On $\mathbb{R}$, $d(x, y) = |x - y|$ is a metric.

Proof. (M1) $|x - y| = 0 \iff x = y$. (M2) $|x - y| = |y - x|$. (M3) triangle inequality for absolute value:

$$|x - z| = |(x - y) + (y - z)| \leq |x - y| + |y - z|.$$

$\square$

Example 2.4 (Discrete metric). On any set X define

$$d_{\text{disc}}(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Then d_{disc} is a metric.

Proof. (M1) holds by definition. (M2) is immediate. For (M3), if $x = z$ then $d(x, z) = 0 \leq d(x, y) + d(y, z)$. If $x \neq z$, then $d(x, z) = 1$; at least one of $x \neq y$ or $y \neq z$ must hold, so $d(x, y) + d(y, z) \geq 1$. $\square$

Example 2.5 (Metrics from norms). Let V be a vector space and $\|\cdot\|$ a norm on V (satisfying positivity, homogeneity, triangle inequality). Then $d(x, y) := \|x - y\|$ defines a metric on V .

Proof. (M1) $\|x - y\| = 0 \iff x - y = 0 \iff x = y$. (M2) $\|x - y\| = \|-(y - x)\| = \|y - x\|$. (M3)

$$\|x - z\| = \|(x - y) + (y - z)\| \leq \|x - y\| + \|y - z\|.$$

$\square$

2.2 Basic inequalities in metric spaces

Lemma 2.6 (Reverse triangle inequality). *For all $x, y, z \in X$,*

$$|d(x, z) - d(y, z)| \leq d(x, y).$$

Proof. By triangle inequality,

$$d(x, z) \leq d(x, y) + d(y, z) \implies d(x, z) - d(y, z) \leq d(x, y).$$

Swapping x and y yields

$$d(y, z) - d(x, z) \leq d(y, x) = d(x, y).$$

Thus $|d(x, z) - d(y, z)| \leq d(x, y)$. $\square$

Theorem 2.7 (Two-point inequality for distances). *For all $x, y, p, q \in X$,*

$$|d(x, y) - d(p, q)| \leq d(x, p) + d(y, q).$$

Proof. Apply triangle inequality in two steps:

$$d(x, y) \leq d(x, p) + d(p, q) + d(q, y) = d(p, q) + d(x, p) + d(y, q),$$

so $d(x, y) - d(p, q) \leq d(x, p) + d(y, q)$. Swap (x, y) and (p, q) to get $d(p, q) - d(x, y) \leq d(x, p) + d(y, q)$. Combine to obtain the absolute value inequality. $\square$

2.3 Constructing new metrics

Definition 2.8 (Equivalent metrics). Two metrics d and ρ on X are *topologically equivalent* if they generate the same open sets (equivalently, the same notion of convergence). A sufficient condition is *bi-Lipschitz equivalence*:

$$\exists c, C > 0 \text{ such that } c d(x, y) \leq \rho(x, y) \leq C d(x, y) \quad \forall x, y.$$

Theorem 2.9 (Bounded transform of a metric). *Let (X, d) be a metric space and define*

$$\tilde{d}(x, y) = \frac{d(x, y)}{1 + d(x, y)}.$$

Then $\tilde{d}$ is a metric on X and $0 \leq \tilde{d}(x, y) < 1$ for all x, y .

Proof. Clearly $0 \leq \tilde{d} < 1$. Symmetry and $\tilde{d}(x, y) = 0 \iff d(x, y) = 0 \iff x = y$ are immediate.

For triangle inequality, set $a = d(x, y)$, $b = d(y, z)$, $c = d(x, z)$ so $c \leq a + b$. It suffices to prove the numeric inequality for all $a, b \geq 0$:

$$\frac{a+b}{1+a+b} \leq \frac{a}{1+a} + \frac{b}{1+b}.$$

Let

$$\Phi(a, b) := \frac{a}{1+a} + \frac{b}{1+b} - \frac{a+b}{1+a+b}.$$

Rewrite each fraction as $1 - \frac{1}{1+a}$ etc.:

$$\Phi(a, b) = \left(1 - \frac{1}{1+a}\right) + \left(1 - \frac{1}{1+b}\right) - \left(1 - \frac{1}{1+a+b}\right) = 1 - \frac{1}{1+a} - \frac{1}{1+b} + \frac{1}{1+a+b}.$$

Thus $\Phi(a, b) \geq 0$ is equivalent to

$$\frac{1}{1+a} + \frac{1}{1+b} \geq 1 + \frac{1}{1+a+b}.$$

Multiply both sides by the positive number $(1+a)(1+b)(1+a+b)$:

$$(1+b)(1+a+b) + (1+a)(1+a+b) \geq (1+a)(1+b)(1+a+b) + (1+a)(1+b).$$

Expand the left:

$$(1+a+b)(2+a+b) = (1+a+b)(a+b+2).$$

Expand the right:

$$(1+a)(1+b)(1+a+b) + (1+a)(1+b) = (1+a)(1+b)((1+a+b)+1) = (1+a)(1+b)(a+b+2).$$

So the inequality becomes

$$(1+a+b)(a+b+2) \geq (1+a)(1+b)(a+b+2).$$

Cancel $(a+b+2) > 0$ to get $1+a+b \geq (1+a)(1+b) = 1+a+b+ab$, i.e. $0 \geq ab$. But $a, b \geq 0$, so $ab \geq 0$ and therefore equality forces $ab = 0$; in general we get

$$1+a+b \leq 1+a+b+ab,$$

so after rearranging we see $\Phi(a, b) = \frac{ab}{(1+a)(1+b)(1+a+b)} \geq 0$. Hence

$$\frac{c}{1+c} \leq \frac{a+b}{1+a+b} \leq \frac{a}{1+a} + \frac{b}{1+b},$$

which is exactly $\tilde{d}(x, z) \leq \tilde{d}(x, y) + \tilde{d}(y, z)$. □

Theorem 2.10 (Pullback metric along an injection). *Let (X, d) be a metric space and let $a : Y \rightarrow X$ be injective. Define*

$$d_Y(y_1, y_2) := d(a(y_1), a(y_2)).$$

Then d_Y is a metric on Y .

Proof. Nonnegativity and symmetry follow from d . Triangle inequality:

$$d_Y(y_1, y_3) = d(a(y_1), a(y_3)) \leq d(a(y_1), a(y_2)) + d(a(y_2), a(y_3)) = d_Y(y_1, y_2) + d_Y(y_2, y_3).$$

If $d_Y(y_1, y_2) = 0$, then $d(a(y_1), a(y_2)) = 0$ so $a(y_1) = a(y_2)$; injectivity implies $y_1 = y_2$. □

Example 2.11 (A useful non-example: pseudo-metric). If we drop (M1) and only require $d(x, x) = 0$, symmetry, triangle inequality, then we get a *pseudometric*. For example, on $X = \mathbb{R}^2$ define $d((x, y), (x', y')) := |x - x'|$. Then distinct points with same x -coordinate have distance 0.

3 Lecture 2: Standard Metric Spaces

3.1 Metrics on $\mathbb{R}^n$ and comparison of norms

Definition 3.1 (Sup metric on $\mathbb{R}^n$). On $\mathbb{R}^n$ define

$$d_\infty(x, y) := \max_{1 \leq i \leq n} |x_i - y_i|.$$

Theorem 3.2 (Sup metric is a metric). d_∞ is a metric on $\mathbb{R}^n$.

Proof. (M1) If $d_\infty(x, y) = 0$, then $\max_i |x_i - y_i| = 0$, hence each $|x_i - y_i| = 0$, so $x = y$. (M2) follows since $|x_i - y_i| = |y_i - x_i|$. (M3) For each i ,

$$|x_i - z_i| \leq |x_i - y_i| + |y_i - z_i| \leq d_\infty(x, y) + d_\infty(y, z).$$

Taking maximum over i gives $d_\infty(x, z) \leq d_\infty(x, y) + d_\infty(y, z)$. $\square$

Theorem 3.3 (Comparison with Euclidean metric). Let $d_2(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$. Then for all $x, y \in \mathbb{R}^n$,

$$d_\infty(x, y) \leq d_2(x, y) \leq \sqrt{n} d_\infty(x, y).$$

Proof. Let $M = d_\infty(x, y) = \max_i |x_i - y_i|$. Then at least one coordinate achieves the max, so

$$d_2(x, y)^2 = \sum_{i=1}^n (x_i - y_i)^2 \geq \max_i (x_i - y_i)^2 = M^2,$$

hence $d_\infty(x, y) = M \leq d_2(x, y)$. Also each $|x_i - y_i| \leq M$, so

$$d_2(x, y)^2 = \sum_{i=1}^n (x_i - y_i)^2 \leq \sum_{i=1}^n M^2 = nM^2,$$

hence $d_2(x, y) \leq \sqrt{n} d_\infty(x, y)$. $\square$

Corollary 3.4 (Same open sets, same convergence on $\mathbb{R}^n$). The metrics d_2 and d_∞ generate the same open sets on $\mathbb{R}^n$.

Proof. The comparison inequalities show d_2 and d_∞ are bi-Lipschitz equivalent, hence yield the same topology. Concretely: for any x and $r > 0$,

$$B_{d_2}(x, r) \subset B_{d_\infty}(x, r) \subset B_{d_2}(x, \sqrt{n}r),$$

so a set is open in one metric iff open in the other. $\square$

3.2 Sequence spaces ℓ_p

Definition 3.5 (ℓ_p). For $1 \leq p < \infty$ define

$$\ell_p := \left\{ x = (x_n)_{n \geq 1} : \sum_{n=1}^{\infty} |x_n|^p < \infty \right\}.$$

Define $\|x\|_p := \left(\sum_{n=1}^{\infty} |x_n|^p \right)^{1/p}$.

Definition 3.6 (Distance on ℓ_p). For $x, y \in \ell_p$, define

$$d_p(x, y) := \|x - y\|_p = \left(\sum_{n=1}^{\infty} |x_n - y_n|^p \right)^{1/p}.$$

3.2.1 Hölder and Minkowski (with proofs)

Theorem 3.7 (Hölder's inequality for finite sums). *Let $1 < p < \infty$ and let q satisfy $\frac{1}{p} + \frac{1}{q} = 1$. For any real numbers a_i, b_i ($i = 1, \dots, m$),*

$$\sum_{i=1}^m |a_i b_i| \leq \left(\sum_{i=1}^m |a_i|^p \right)^{1/p} \left(\sum_{i=1}^m |b_i|^q \right)^{1/q}.$$

Proof. If either side is 0, done. Assume $\sum |a_i|^p > 0$ and $\sum |b_i|^q > 0$. Define

$$A := \left(\sum_{i=1}^m |a_i|^p \right)^{1/p}, \quad B := \left(\sum_{i=1}^m |b_i|^q \right)^{1/q},$$

and set $\alpha_i := |a_i|/A$, $\beta_i := |b_i|/B$. Then $\sum \alpha_i^p = 1$ and $\sum \beta_i^q = 1$.

We use Young's inequality: for $u, v \geq 0$,

$$uv \leq \frac{u^p}{p} + \frac{v^q}{q}.$$

(Proof: consider $\phi(u) = \frac{u^p}{p} - uv + \frac{v^q}{q}$; its minimum occurs at $u = v^{q-1}$ and equals 0.)

Apply Young to $u = \alpha_i$, $v = \beta_i$:

$$\alpha_i \beta_i \leq \frac{\alpha_i^p}{p} + \frac{\beta_i^q}{q}.$$

Sum over i :

$$\sum_{i=1}^m \alpha_i \beta_i \leq \frac{1}{p} \sum \alpha_i^p + \frac{1}{q} \sum \beta_i^q = \frac{1}{p} \cdot 1 + \frac{1}{q} \cdot 1 = 1.$$

Now multiply by AB :

$$\sum_{i=1}^m |a_i b_i| = AB \sum \alpha_i \beta_i \leq AB,$$

which is exactly Hölder. □

Theorem 3.8 (Minkowski inequality for finite sums). *Let $1 \leq p < \infty$. For real numbers u_i, v_i ($i = 1, \dots, m$),*

$$\left(\sum_{i=1}^m |u_i + v_i|^p \right)^{1/p} \leq \left(\sum_{i=1}^m |u_i|^p \right)^{1/p} + \left(\sum_{i=1}^m |v_i|^p \right)^{1/p}.$$

Proof. For $p = 1$ it is the ordinary triangle inequality for sums. Assume $1 < p < \infty$ and let q be conjugate to p .

Let $s_i := |u_i + v_i|$ and $S := (\sum s_i^p)^{1/p}$. If $S = 0$, then all $s_i = 0$ and the inequality holds. Assume $S > 0$. Then

$$S^p = \sum s_i^p = \sum s_i^{p-1} s_i = \sum s_i^{p-1} |u_i + v_i| \leq \sum s_i^{p-1} |u_i| + \sum s_i^{p-1} |v_i|.$$

Apply Hölder to the first sum with sequences (s_i^{p-1}) and $(|u_i|)$:

$$\sum s_i^{p-1} |u_i| \leq \left(\sum (s_i^{p-1})^q \right)^{1/q} \left(\sum |u_i|^p \right)^{1/p}.$$

But $(p-1)q = p$ (since $q = \frac{p}{p-1}$), so

$$\sum (s_i^{p-1})^q = \sum s_i^{(p-1)q} = \sum s_i^p = S^p.$$

Hence $(\sum (s_i^{p-1})^q)^{1/q} = (S^p)^{1/q} = S^{p/q} = S^{p-1}$ (because $p/q = p-1$). Thus

$$\sum s_i^{p-1} |u_i| \leq S^{p-1} \left(\sum |u_i|^p \right)^{1/p}.$$

Similarly,

$$\sum s_i^{p-1} |v_i| \leq S^{p-1} \left(\sum |v_i|^p \right)^{1/p}.$$

Combine:

$$S^p \leq S^{p-1} \left(\sum |u_i|^p \right)^{1/p} + S^{p-1} \left(\sum |v_i|^p \right)^{1/p}.$$

Divide by $S^{p-1} > 0$:

$$S \leq \left(\sum |u_i|^p \right)^{1/p} + \left(\sum |v_i|^p \right)^{1/p}.$$

□

Theorem 3.9 (Minkowski for series). *If $u = (u_n)$ and $v = (v_n)$ are sequences with $\sum |u_n|^p < \infty$ and $\sum |v_n|^p < \infty$, then*

$$\left(\sum_{n=1}^{\infty} |u_n + v_n|^p \right)^{1/p} \leq \left(\sum_{n=1}^{\infty} |u_n|^p \right)^{1/p} + \left(\sum_{n=1}^{\infty} |v_n|^p \right)^{1/p}.$$

Proof. Apply finite Minkowski to partial sums:

$$\left(\sum_{n=1}^m |u_n + v_n|^p \right)^{1/p} \leq \left(\sum_{n=1}^m |u_n|^p \right)^{1/p} + \left(\sum_{n=1}^m |v_n|^p \right)^{1/p}.$$

As $m \rightarrow \infty$, the right-hand side increases to $\|u\|_p + \|v\|_p$ and the left-hand side increases to $\|u + v\|_p$ by monotone convergence of partial sums. □

Theorem 3.10 (Metric property of d_p on ℓ_p). *For $1 \leq p < \infty$, the function $d_p(x, y) = \|x - y\|_p$ defines a metric on ℓ_p .*

Proof. (M2) symmetry is clear. (M1) If $d_p(x, y) = 0$, then $\sum |x_n - y_n|^p = 0$ so each $|x_n - y_n|^p = 0$, hence $x_n = y_n$ for all n . (M3) By Minkowski for series with $u = x - y$ and $v = y - z$, we get

$$\|x - z\|_p = \|(x - y) + (y - z)\|_p \leq \|x - y\|_p + \|y - z\|_p,$$

i.e. $d_p(x, z) \leq d_p(x, y) + d_p(y, z)$. □

4 Lecture 3: Topology Induced by a Metric

4.1 Open sets, closed sets

Definition 4.1 (Open set). A set $G \subset X$ is *open* if for every $x \in G$ there exists $r > 0$ such that $B(x, r) \subset G$.

Definition 4.2 (Closed set). A set $F \subset X$ is *closed* if $X \setminus F$ is open.

Theorem 4.3 (Unions and intersections of open sets). *Let (X, d) be a metric space.*

(a) *Arbitrary unions of open sets are open.*

(b) *Finite intersections of open sets are open.*

Proof. (a) Let $G = \cup_{\alpha \in I} G_\alpha$ with each G_α open. If $x \in G$, then $x \in G_{\alpha_0}$ for some α_0 . Since G_{α_0} is open, $\exists r > 0$ with $B(x, r) \subset G_{\alpha_0} \subset G$. Hence G is open.

(b) Let $G = \cap_{k=1}^n G_k$ with each G_k open. Fix $x \in G$. Then for each k there exists $r_k > 0$ with $B(x, r_k) \subset G_k$. Let $r := \min\{r_1, \dots, r_n\} > 0$. Then $B(x, r) \subset B(x, r_k) \subset G_k$ for all k , hence $B(x, r) \subset \cap_{k=1}^n G_k = G$. So G is open. $\square$

Example 4.4 (Infinite intersections need not be open). In $(\mathbb{R}, |\cdot|)$, each $G_n = (-1/n, 1/n)$ is open, but

$$\bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\},$$

which is not open in $\mathbb{R}$.

Theorem 4.5 (Closed sets: unions/intersections). (a) *Arbitrary intersections of closed sets are closed.*

(b) *Finite unions of closed sets are closed.*

Proof. Use complements and the previous theorem, together with De Morgan's laws:

$$X \setminus \left(\bigcap_{\alpha} F_{\alpha} \right) = \bigcup_{\alpha} (X \setminus F_{\alpha}), \quad X \setminus (F_1 \cup F_2) = (X \setminus F_1) \cap (X \setminus F_2).$$

$\square$

4.2 Closure, interior, boundary

Definition 4.6 (Interior). For $A \subset X$, the *interior* $\text{Int}(A)$ is the set of all points $x \in A$ for which some ball $B(x, r)$ lies in A .

Definition 4.7 (Closure via limit points of balls). For $A \subset X$, the *closure* $\overline{A}$ is the set of all $x \in X$ such that every ball around x meets A :

$$x \in \overline{A} \iff \forall r > 0, B(x, r) \cap A \neq \emptyset.$$

Why this equals the intersection-of-closed-sets definition. Let $\mathcal{F}$ be the family of all closed sets containing A and let $C = \cap_{F \in \mathcal{F}} F$. If $x \notin C$, then there exists a closed $F \supset A$ with $x \notin F$, so $x \in X \setminus F$, which is open. Hence there is $r > 0$ with $B(x, r) \subset X \setminus F$, so $B(x, r) \cap A = \emptyset$ (since $A \subset F$). Conversely, if $B(x, r) \cap A = \emptyset$ for some r , then $X \setminus B(x, r)$ is closed and contains A but not x , hence $x \notin C$. Thus $x \in C$ iff every ball around x meets A . $\square$

Definition 4.8 (Boundary). For $A \subset X$, the *boundary* is

$$\partial A := \overline{A} \setminus \text{Int}(A).$$

Definition 4.9 (Limit point and derived set). A point $x \in X$ is a *limit point* (or *accumulation point*) of A if every ball around x meets $A \setminus \{x\}$. The set of all limit points is the *derived set* A' .

Proposition 4.10 (Basic relations). *For any $A \subset X$:*

(a) $\text{Int}(A) \subset A \subset \overline{A}$.

(b) A is open $\iff A = \text{Int}(A)$.

(c) A is closed $\iff A = \overline{A}$.

(d) ∂A is closed.

Proof. (a) If $x \in \text{Int}(A)$ then $x \in A$ by definition. If $x \in A$, then every ball around x meets A (it contains x), so $x \in \bar{A}$.

(b) If A is open, every $x \in A$ has a ball contained in A , so $x \in \text{Int}(A)$ and $A \subset \text{Int}(A)$. Together with $\text{Int}(A) \subset A$ gives equality. Conversely, if $A = \text{Int}(A)$ then every $x \in A$ is an interior point, so A is open.

(c) If A is closed and $A \subset \bar{A}$, but $\bar{A}$ is the *smallest* closed set containing A , hence $\bar{A} \subset A$ and equality holds. Conversely, if $A = \bar{A}$, then A equals a closure and is therefore closed.

(d) Since $\bar{A}$ is closed and $\text{Int}(A)$ is open, its complement $X \setminus \text{Int}(A)$ is closed. Then $\partial A = \bar{A} \cap (X \setminus \text{Int}(A))$ is intersection of closed sets, hence closed. $\square$

Theorem 4.11 (Closure of a union). *For all $A, B \subset X$,*

$$\overline{A \cup B} = \bar{A} \cup \bar{B}.$$

Proof. First, $A \subset \bar{A}$ and $B \subset \bar{B}$ implies $A \cup B \subset \bar{A} \cup \bar{B}$. The right side is closed (finite union of closed sets), so it contains the smallest closed set containing $A \cup B$, namely $\overline{A \cup B}$:

$$\overline{A \cup B} \subset \bar{A} \cup \bar{B}.$$

Conversely, since $A \subset A \cup B$, by monotonicity of closure we have $\bar{A} \subset \overline{A \cup B}$; similarly $\bar{B} \subset \overline{A \cup B}$. Thus $\bar{A} \cup \bar{B} \subset \overline{A \cup B}$. $\square$

Theorem 4.12 (Open sets are unions of balls). *A subset $G \subset X$ is open iff it can be written as a union of open balls.*

Proof. If G is open, for each $x \in G$ choose $r_x > 0$ with $B(x, r_x) \subset G$. Then clearly $G = \cup_{x \in G} B(x, r_x)$. Conversely, each ball is open, and arbitrary unions of open sets are open. $\square$

Theorem 4.13 (Closed balls are closed). *For $x \in X$ and $r \geq 0$, the closed ball $\bar{B}(x, r) = \{y : d(x, y) \leq r\}$ is closed.*

Proof. Let $y \notin \bar{B}(x, r)$, so $d(x, y) > r$. Set $\eta := d(x, y) - r > 0$. If $d(y, z) < \eta$, then by triangle inequality,

$$d(x, z) \geq d(x, y) - d(y, z) > d(x, y) - \eta = r,$$

so $z \notin \bar{B}(x, r)$. Therefore $B(y, \eta) \subset X \setminus \bar{B}(x, r)$, so the complement is open and $\bar{B}(x, r)$ is closed. $\square$

5 Lecture 4: Density and Examples in $\mathbb{R}$ and $\mathbb{R}^2$

Definition 5.1 (Dense subset). A subset $A \subset X$ is *dense* in X if $\bar{A} = X$. Equivalently: every nonempty open set in X intersects A .

Theorem 5.2 ($\mathbb{Q}$ is dense in $\mathbb{R}$). *In $(\mathbb{R}, |\cdot|)$, the rationals $\mathbb{Q}$ are dense.*

Proof. Let (a, b) be any nonempty open interval. Choose $n \in \mathbb{N}$ with $\frac{1}{n} < b - a$. Let $k = \lceil an \rceil$ (smallest integer $\geq an$). Then $\frac{k}{n} \geq a$ and

$$\frac{k}{n} < a + \frac{1}{n} < b.$$

Hence $\frac{k}{n} \in (a, b) \cap \mathbb{Q}$, proving density. $\square$

Theorem 5.3 ($\mathbb{Q}^2$ is dense in $\mathbb{R}^2$). *Let $A = \{(p, q) : p, q \in \mathbb{Q}\} \subset \mathbb{R}^2$ with the Euclidean metric. Then A is dense in $\mathbb{R}^2$.*

Proof. Fix $(x, y) \in \mathbb{R}^2$ and $\varepsilon > 0$. Choose rationals p, q such that $|p - x| < \varepsilon/\sqrt{2}$ and $|q - y| < \varepsilon/\sqrt{2}$ (density of $\mathbb{Q}$ in $\mathbb{R}$). Then

$$\sqrt{(p - x)^2 + (q - y)^2} < \sqrt{\varepsilon^2/2 + \varepsilon^2/2} = \varepsilon,$$

so every ball around (x, y) meets A . □

Example 5.4 (Compute interior/closure/boundary in $\mathbb{R}$). Let $A = (1, 2] \cup (3, 4) \subset \mathbb{R}$. Then

$$\text{Int}(A) = (1, 2) \cup (3, 4), \quad \overline{A} = [1, 2] \cup [3, 4], \quad \partial A = \{1, 2, 3, 4\}.$$

Also $A' = [1, 2] \cup [3, 4]$.

Proof. Interior points are those not at the “one-sided” endpoints. Closure adds endpoints. Boundary is closure minus interior. Every point in the closed intervals is a limit point because every neighborhood contains infinitely many points of the interval portion of A distinct from the point. □

Theorem 5.5 (Interior of unions (and strictness)). *For all $A, B \subset X$,*

$$\text{Int}(A) \cup \text{Int}(B) \subset \text{Int}(A \cup B),$$

and equality can fail.

Proof. If $x \in \text{Int}(A)$, then $\exists r > 0$ with $B(x, r) \subset A \subset A \cup B$, hence $x \in \text{Int}(A \cup B)$. Same for $\text{Int}(B)$, so the inclusion holds.

For strictness in $\mathbb{R}$, take $A = \mathbb{Q}$ and $B = \mathbb{R} \setminus \mathbb{Q}$. Then $\text{Int}(A) = \text{Int}(B) = \emptyset$ (every interval contains both rationals and irrationals), but $A \cup B = \mathbb{R}$, hence $\text{Int}(A \cup B) = \mathbb{R}$. □

6 Lecture 5: Sequences, Limits, and Cauchy Sequences

6.1 Definitions

Definition 6.1 (Convergence of sequences). A sequence (x_n) in (X, d) *converges* to $x \in X$, written $x_n \rightarrow x$, if

$$\forall \varepsilon > 0 \exists N \text{ such that } n \geq N \Rightarrow d(x_n, x) < \varepsilon.$$

Definition 6.2 (Cauchy sequence). A sequence (x_n) is *Cauchy* if

$$\forall \varepsilon > 0 \exists N \text{ such that } m, n \geq N \Rightarrow d(x_m, x_n) < \varepsilon.$$

Lemma 6.3 (Uniqueness of limits). *If $x_n \rightarrow x$ and $x_n \rightarrow y$ in a metric space, then $x = y$.*

Proof. Assume $x \neq y$. Let $\eta := d(x, y) > 0$. Choose N_1 such that $d(x_n, x) < \eta/3$ for $n \geq N_1$ and N_2 such that $d(x_n, y) < \eta/3$ for $n \geq N_2$. For $n \geq \max\{N_1, N_2\}$,

$$d(x, y) \leq d(x, x_n) + d(x_n, y) < \eta/3 + \eta/3 = 2\eta/3,$$

contradiction. Hence $x = y$. □

Theorem 6.4 (Convergent $\Rightarrow$ Cauchy). *Every convergent sequence in a metric space is Cauchy.*

Proof. Assume $x_n \rightarrow x$. Fix $\varepsilon > 0$ and choose N such that $d(x_n, x) < \varepsilon/2$ for $n \geq N$. Then for $m, n \geq N$,

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

□

Theorem 6.5 (Continuity of the distance function along sequences). *If $x_n \rightarrow x$ and $y_n \rightarrow y$ in (X, d) , then*

$$d(x_n, y_n) \rightarrow d(x, y).$$

Proof. Use the two-point inequality:

$$|d(x_n, y_n) - d(x, y)| \leq d(x_n, x) + d(y_n, y) \xrightarrow{n \rightarrow \infty} 0.$$

□

Example 6.6 (A Cauchy sequence in $\mathbb{Q}$ that does not converge in $\mathbb{Q}$). In $(\mathbb{Q}, |\cdot|)$, the sequence $a_n = (1 + \frac{1}{n})^n$ is Cauchy but does not converge in $\mathbb{Q}$.

Proof. In $\mathbb{R}$, $a_n \rightarrow e$. Hence (a_n) is Cauchy in $\mathbb{R}$. The metric on $\mathbb{Q}$ is the restriction of $|\cdot|$, so the Cauchy condition is the same; thus (a_n) is Cauchy in $\mathbb{Q}$. If it converged in $\mathbb{Q}$ to q , it would converge in $\mathbb{R}$ to q by uniqueness of limits; but the real limit is $e \notin \mathbb{Q}$. □

7 Lecture 6: Completeness

Definition 7.1 (Complete metric space). A metric space (X, d) is *complete* if every Cauchy sequence in X converges to a point of X .

Example 7.2 (Incomplete metric space). $(\mathbb{Q}, |\cdot|)$ is not complete (previous example).

7.1 Completeness of ℓ_p (full proof)

Theorem 7.3 (ℓ_p is complete). *For $1 \leq p < \infty$, the metric space (ℓ_p, d_p) is complete.*

Proof. Let $(x^{(k)})_{k \geq 1}$ be a Cauchy sequence in ℓ_p , where $x^{(k)} = (x_n^{(k)})_{n \geq 1}$. We prove there exists $x \in \ell_p$ such that $d_p(x^{(k)}, x) \rightarrow 0$.

Step 1: coordinatewise convergence. Fix $n \in \mathbb{N}$. For any k, m ,

$$|x_n^{(k)} - x_n^{(m)}|^p \leq \sum_{j=1}^{\infty} |x_j^{(k)} - x_j^{(m)}|^p = d_p(x^{(k)}, x^{(m)})^p.$$

Since $(x^{(k)})$ is Cauchy, the right-hand side $\rightarrow 0$ as $k, m \rightarrow \infty$, so $(x_n^{(k)})_k$ is Cauchy in $\mathbb{R}$. Therefore it converges to some real number x_n . Define $x = (x_n)_{n \geq 1}$.

Step 2: show $x \in \ell_p$. Because $(x^{(k)})$ is Cauchy, it is bounded in ℓ_p -norm. Indeed, choose K such that $d_p(x^{(k)}, x^{(K)}) \leq 1$ for all $k \geq K$. Then for $k \geq K$,

$$\|x^{(k)}\|_p \leq \|x^{(K)}\|_p + \|x^{(k)} - x^{(K)}\|_p \leq \|x^{(K)}\|_p + 1 =: M.$$

So $\|x^{(k)}\|_p^p = \sum_{n=1}^{\infty} |x_n^{(k)}|^p \leq M^p$ for all $k \geq K$.

Fix N . By coordinatewise convergence $x_n^{(k)} \rightarrow x_n$ for each $n \leq N$. Hence for each fixed N ,

$$\sum_{n=1}^N |x_n|^p = \sum_{n=1}^N \lim_{k \rightarrow \infty} |x_n^{(k)}|^p = \lim_{k \rightarrow \infty} \sum_{n=1}^N |x_n^{(k)}|^p \leq \limsup_{k \rightarrow \infty} \sum_{n=1}^{\infty} |x_n^{(k)}|^p \leq M^p.$$

The sequence of partial sums $\sum_{n=1}^N |x_n|^p$ is increasing in N and bounded above by M^p , hence converges to a finite limit:

$$\sum_{n=1}^{\infty} |x_n|^p < \infty.$$

Thus $x \in \ell_p$.

Step 3: show $x^{(k)} \rightarrow x$ in ℓ_p . Fix $\varepsilon > 0$. Choose K such that $d_p(x^{(k)}, x^{(m)}) < \varepsilon$ for all $k, m \geq K$. Fix $k \geq K$. For each N ,

$$\left(\sum_{n=1}^N |x_n^{(k)} - x_n| \right)^{1/p} = \left(\sum_{n=1}^N \lim_{m \rightarrow \infty} |x_n^{(k)} - x_n^{(m)}|^p \right)^{1/p} = \lim_{m \rightarrow \infty} \left(\sum_{n=1}^N |x_n^{(k)} - x_n^{(m)}|^p \right)^{1/p}.$$

(The last equality uses continuity of finite sums and $t \mapsto t^{1/p}$.) Now for every $m \geq K$,

$$\left(\sum_{n=1}^N |x_n^{(k)} - x_n^{(m)}|^p \right)^{1/p} \leq \left(\sum_{n=1}^{\infty} |x_n^{(k)} - x_n^{(m)}|^p \right)^{1/p} = d_p(x^{(k)}, x^{(m)}) < \varepsilon.$$

Taking $m \rightarrow \infty$ yields

$$\left(\sum_{n=1}^N |x_n^{(k)} - x_n|^p \right)^{1/p} \leq \varepsilon \quad \text{for all } N.$$

Letting $N \rightarrow \infty$ (monotone convergence of partial sums) gives $d_p(x^{(k)}, x) \leq \varepsilon$ for all $k \geq K$. Therefore $x^{(k)} \rightarrow x$, proving completeness. $\square$

7.2 Closed subsets and completeness

Theorem 7.4 (Closed subspace of a complete space is complete). *Let (X, d) be complete and let $Y \subset X$ be closed (with the subspace metric). Then Y is complete.*

Proof. Let (y_n) be Cauchy in Y . Then it is Cauchy in X (same distances), so by completeness of X it converges to some $x \in X$. Since Y is closed and $y_n \in Y$ with $y_n \rightarrow x$, we must have $x \in Y$. Thus (y_n) converges in Y . $\square$

Theorem 7.5 (Complete subspace of a metric space is closed (in a complete ambient space)). *Let (X, d) be complete and $Y \subset X$. If Y is complete (with subspace metric), then Y is closed in X .*

Proof. Take $x \in \overline{Y}$. Then there exists a sequence $(y_n) \subset Y$ with $y_n \rightarrow x$ in X . Every convergent sequence is Cauchy, so (y_n) is Cauchy in Y . Completeness of Y implies $y_n \rightarrow y$ for some $y \in Y$. But limits are unique in X , so $x = y \in Y$. Hence $\overline{Y} \subset Y$, so Y is closed. $\square$

Corollary 7.6 (Closed $\iff$ complete subspace (inside a complete space)). *If (X, d) is complete, then for any $Y \subset X$,*

$$Y \text{ is closed in } X \iff Y \text{ is complete (subspace metric)}.$$

8 Lecture 7: Continuity and Uniform Continuity

8.1 Continuity

Definition 8.1 (Continuity at a point). Let (X, d_X) , (Y, d_Y) be metric spaces and $f : X \rightarrow Y$. We say f is *continuous at $x_0 \in X$* if

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ such that } d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon.$$

We say f is *continuous on X* if it is continuous at every $x_0 \in X$.

Theorem 8.2 (Sequential characterization of continuity). *f is continuous at x_0 iff for every sequence $x_n \rightarrow x_0$ we have $f(x_n) \rightarrow f(x_0)$.*

Proof. ($\Rightarrow$) Fix $\varepsilon > 0$ and let δ be given by continuity at x_0 . Since $x_n \rightarrow x_0$, there exists N such that $n \geq N \Rightarrow d_X(x_n, x_0) < \delta$. Then for $n \geq N$, $d_Y(f(x_n), f(x_0)) < \varepsilon$, so $f(x_n) \rightarrow f(x_0)$.

($\Leftarrow$) Contrapositive. Assume f is not continuous at x_0 . Then $\exists \varepsilon_0 > 0$ such that for all $\delta > 0$ there exists x with $d_X(x, x_0) < \delta$ but $d_Y(f(x), f(x_0)) \geq \varepsilon_0$. Choose x_n with $\delta = 1/n$. Then $x_n \rightarrow x_0$ but $f(x_n)$ does not converge to $f(x_0)$ (distance $\geq \varepsilon_0$ infinitely often). This contradicts the sequential property. Hence f must be continuous at x_0 . $\square$

Theorem 8.3 (Open set preimage criterion). *$f : X \rightarrow Y$ is continuous on X iff for every open $G \subset Y$, the preimage $f^{-1}(G)$ is open in X .*

Proof. ($\Rightarrow$) Let $G \subset Y$ be open and $x \in f^{-1}(G)$. Then $f(x) \in G$, so there exists $\varepsilon > 0$ with $B_Y(f(x), \varepsilon) \subset G$. By continuity at x , there exists $\delta > 0$ such that $d_X(x', x) < \delta \Rightarrow d_Y(f(x'), f(x)) < \varepsilon$, hence $f(x') \in B_Y(f(x), \varepsilon) \subset G$. So $x' \in f^{-1}(G)$ whenever $d_X(x', x) < \delta$, i.e. $B_X(x, \delta) \subset f^{-1}(G)$. Thus $f^{-1}(G)$ is open.

($\Leftarrow$) Fix $x_0 \in X$ and $\varepsilon > 0$. The ball $B_Y(f(x_0), \varepsilon)$ is open in Y . Hence $U := f^{-1}(B_Y(f(x_0), \varepsilon))$ is open in X and contains x_0 . So there exists $\delta > 0$ with $B_X(x_0, \delta) \subset U$. Therefore if $d_X(x, x_0) < \delta$, then $x \in U$ so $f(x) \in B_Y(f(x_0), \varepsilon)$, i.e. $d_Y(f(x), f(x_0)) < \varepsilon$. Thus f is continuous at x_0 . $\square$

Example 8.4 (Every function from a discrete metric space is continuous). If (X, d_{disc}) is discrete and $f : X \rightarrow Y$ is any function, then f is continuous.

Proof. In a discrete metric space, every subset is open: for any $A \subset X$ and any $x \in A$, $B(x, 1/2) = \{x\} \subset A$. Hence preimages of open sets are automatically open, so f is continuous by the preimage criterion. $\square$

8.2 Uniform continuity

Definition 8.5 (Uniform continuity). *$f : X \rightarrow Y$ is uniformly continuous if*

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ such that } d_X(x, x') < \delta \Rightarrow d_Y(f(x), f(x')) < \varepsilon \text{ for all } x, x' \in X.$$

Proposition 8.6 (Lipschitz $\Rightarrow$ uniformly continuous). *If $\exists L \geq 0$ such that $d_Y(f(x), f(x')) \leq L d_X(x, x')$ for all x, x' , then f is uniformly continuous.*

Proof. Given $\varepsilon > 0$, take $\delta = \varepsilon/L$ if $L > 0$ (and any δ if $L = 0$). Then $d_X(x, x') < \delta \Rightarrow d_Y(f(x), f(x')) < \varepsilon$. $\square$

Theorem 8.7 (Uniform continuity sends Cauchy sequences to Cauchy sequences). *If $f : X \rightarrow Y$ is uniformly continuous and (x_n) is Cauchy in X , then $(f(x_n))$ is Cauchy in Y .*

Proof. Let $\varepsilon > 0$ and choose $\delta > 0$ given by uniform continuity. Since (x_n) is Cauchy, there exists N such that $m, n \geq N \Rightarrow d_X(x_m, x_n) < \delta$. Then for $m, n \geq N$ we have $d_Y(f(x_m), f(x_n)) < \varepsilon$. Thus $(f(x_n))$ is Cauchy. $\square$

Theorem 8.8 (Distance to a set is 1-Lipschitz). *Let $A \subset X$ be nonempty and define $g(x) = \text{dist}(x, A) := \inf_{a \in A} d(x, a)$. Then for all $x, y \in X$,*

$$|g(x) - g(y)| \leq d(x, y).$$

In particular g is uniformly continuous.

Proof. Fix x, y . For any $a \in A$, triangle inequality gives $d(x, a) \leq d(x, y) + d(y, a)$. Taking infimum over $a \in A$ yields

$$g(x) = \inf_{a \in A} d(x, a) \leq d(x, y) + \inf_{a \in A} d(y, a) = d(x, y) + g(y).$$

So $g(x) - g(y) \leq d(x, y)$. Swapping x, y gives $g(y) - g(x) \leq d(x, y)$. Therefore $|g(x) - g(y)| \leq d(x, y)$. $\square$

Theorem 8.9 (Heine–Cantor). *If $K \subset X$ is compact and $f : K \rightarrow Y$ is continuous, then f is uniformly continuous on K .*

Proof. Assume f is not uniformly continuous. Then $\exists \varepsilon_0 > 0$ and sequences $x_n, y_n \in K$ such that $d_X(x_n, y_n) \rightarrow 0$ but $d_Y(f(x_n), f(y_n)) \geq \varepsilon_0$ for all n . By compactness, (x_n) has a convergent subsequence $x_{n_k} \rightarrow x \in K$. Since $d_X(x_{n_k}, y_{n_k}) \rightarrow 0$, we also have $y_{n_k} \rightarrow x$. Continuity implies $f(x_{n_k}) \rightarrow f(x)$ and $f(y_{n_k}) \rightarrow f(x)$, hence $d_Y(f(x_{n_k}), f(y_{n_k})) \rightarrow 0$, contradicting $d_Y(f(x_{n_k}), f(y_{n_k})) \geq \varepsilon_0$. $\square$

9 Lecture 8: Compactness, Total Boundedness, Sequential Compactness

9.1 Compactness

Definition 9.1 (Open cover and compactness). Let $K \subset X$. A family of open sets $\{U_\alpha\}_{\alpha \in I}$ is an *open cover* of K if

$$K \subset \bigcup_{\alpha \in I} U_\alpha.$$

The set K is *compact* if every open cover of K has a finite subcover. A space X is compact if X is compact as a subset of itself.

Definition 9.2 (Sequential compactness). A metric space X is *sequentially compact* if every sequence in X has a convergent subsequence.

Definition 9.3 (Total boundedness). X is *totally bounded* if for every $\varepsilon > 0$ there exist finitely many points $x_1, \dots, x_m \in X$ such that

$$X \subset \bigcup_{i=1}^m B(x_i, \varepsilon).$$

Theorem 9.4 (Compact $\Rightarrow$ totally bounded). *If X is compact, then X is totally bounded.*

Proof. Fix $\varepsilon > 0$. The family $\{B(x, \varepsilon)\}_{x \in X}$ is an open cover of X . By compactness, choose a finite subcover $X \subset \bigcup_{i=1}^m B(x_i, \varepsilon)$. $\square$

Theorem 9.5 (Compact $\Rightarrow$ sequentially compact (detailed)). *If X is compact metric space, then X is sequentially compact.*

Proof. Let (x_n) be a sequence in X . We will build a convergent subsequence.

By compactness $\Rightarrow$ totally bounded, cover X by finitely many balls of radius 1. At least one of these balls contains infinitely many terms of the sequence (pigeonhole principle). Choose a subsequence $(x_n^{(1)})$ of (x_n) contained entirely in some ball B_1 of radius 1.

Now cover B_1 (hence X) by finitely many balls of radius $1/2$. One of them contains infinitely many terms of $(x_n^{(1)})$; choose a further subsequence $(x_n^{(2)})$ contained in a ball B_2 of radius $1/2$.

Continue inductively: having chosen a subsequence $(x_n^{(k)})$ contained in a ball B_k of radius $2^{-(k-1)}$, cover B_k by finitely many balls of radius 2^{-k} and choose a subsequence $(x_n^{(k+1)})$ contained in one such ball B_{k+1} .

Thus we obtain nested subsequences

$$(x_n) \supset (x_n^{(1)}) \supset (x_n^{(2)}) \supset \dots$$

and nested balls $B_1 \supset B_2 \supset \dots$ with $\text{radius}(B_k) = 2^{-(k-1)}$ and $(x_n^{(k)}) \subset B_k$.

Define the diagonal subsequence $y_k := x_k^{(k)}$ (the k -th term of the k -th subsequence). Then for $m, n \geq k$, both y_m and y_n lie in B_k , so

$$d(y_m, y_n) \leq \text{diam}(B_k) \leq 2 \cdot 2^{-(k-1)} = 2^{-(k-2)} \xrightarrow[k \rightarrow \infty]{} 0.$$

Hence (y_k) is Cauchy.

It remains to show (y_k) converges. In a compact metric space, every Cauchy sequence converges: since X is sequentially compact implies complete, but we can prove directly: the sequence (y_k) has a convergent subsequence (by compactness again), and a Cauchy sequence with a convergent subsequence must converge to the same limit. Indeed, let $y_{k_j} \rightarrow y$. Fix $\varepsilon > 0$. Choose K such that $m, n \geq K \Rightarrow d(y_m, y_n) < \varepsilon/2$ (Cauchy). Also choose j such that $k_j \geq K$ and $d(y_{k_j}, y) < \varepsilon/2$. Then for any $n \geq K$,

$$d(y_n, y) \leq d(y_n, y_{k_j}) + d(y_{k_j}, y) < \varepsilon/2 + \varepsilon/2 = \varepsilon,$$

so $y_n \rightarrow y$. Thus (y_k) converges, hence (x_n) has a convergent subsequence. $\square$

Theorem 9.6 (Compact sets are closed and bounded). *If $K \subset X$ is compact, then K is closed and bounded.*

Proof. Bounded: Fix $x_0 \in X$. The open balls $\{B(x_0, n)\}_{n \in \mathbb{N}}$ cover K because for each $x \in K$, $d(x, x_0) < n$ for some n (take $n > d(x, x_0)$). By compactness, $K \subset B(x_0, N)$ for some N , so K is bounded.

Closed: Let $(x_n) \subset K$ with $x_n \rightarrow x$ in X . By compactness, (x_n) has a convergent subsequence $x_{n_k} \rightarrow y \in K$. But $x_{n_k} \rightarrow x$ as well (subsequence of a convergent sequence). By uniqueness of limits, $x = y \in K$. Hence K is closed. $\square$

Theorem 9.7 (Closed subset of compact is compact). *If K is compact and $F \subset K$ is closed (in the subspace topology), then F is compact.*

Proof. Let $\{U_\alpha\}$ be an open cover of F (each U_α open in K , hence $U_\alpha = K \cap V_\alpha$ with V_α open in X). Then $\{U_\alpha\} \cup \{K \setminus F\}$ is an open cover of K . Compactness gives a finite subcover $U_{\alpha_1}, \dots, U_{\alpha_m}, K \setminus F$. Removing $K \setminus F$ leaves a finite subcover of F . $\square$

9.2 Sequential compactness $\Leftrightarrow$ completeness + total boundedness

Theorem 9.8 (Characterization). *For a metric space X , the following are equivalent:*

- (1) X is sequentially compact.
- (2) X is complete and totally bounded.

Proof. (1) $\Rightarrow$ (2): Assume sequentially compact.

Completeness: Let (x_n) be Cauchy. By sequential compactness, it has a convergent subsequence $x_{n_k} \rightarrow x$. We claim $x_n \rightarrow x$. Fix $\varepsilon > 0$. Choose N with $m, n \geq N \Rightarrow d(x_m, x_n) < \varepsilon/2$ (Cauchy). Choose k so that $n_k \geq N$ and $d(x_{n_k}, x) < \varepsilon/2$. Then for $n \geq N$,

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

So $x_n \rightarrow x$ and X is complete.

Total boundedness: Suppose X is not totally bounded. Then $\exists \varepsilon > 0$ such that no finite union of ε -balls covers X . Construct a sequence (x_n) inductively: choose $x_1 \in X$. Having chosen $x_1, \dots, x_n$, since $\cup_{k=1}^n B(x_k, \varepsilon)$ does not cover X , choose $x_{n+1} \in X \setminus \cup_{k=1}^n B(x_k, \varepsilon)$. Then for $m \neq n$, we have $d(x_m, x_n) \geq \varepsilon$. Such a sequence cannot have a Cauchy subsequence, hence cannot have a convergent subsequence, contradicting sequential compactness. Thus X is totally bounded.

(2) $\Rightarrow$ (1): Assume X is complete and totally bounded. Let (x_n) be any sequence. We build a Cauchy subsequence using total boundedness.

Cover X by finitely many balls of radius 1. One ball contains infinitely many x_n ; choose a subsequence inside it. Cover that ball by finitely many balls of radius $1/2$; choose a further subsequence inside one such ball. Continue to get nested subsequences with diameters going to 0 exactly as in the compact $\Rightarrow$ sequentially compact proof. The diagonal subsequence is Cauchy. By completeness it converges. Hence every sequence has a convergent subsequence, so X is sequentially compact. $\square$

10 Lecture 9: Separability

Definition 10.1 (Separable metric space). A metric space X is *separable* if it has a countable dense subset.

Theorem 10.2 (Totally bounded $\Rightarrow$ separable). *Every totally bounded metric space is separable.*

Proof. For each $n \in \mathbb{N}$, cover X by finitely many balls of radius $1/n$:

$$X \subset \bigcup_{i=1}^{m_n} B(x_{n,i}, 1/n).$$

Let $D = \{x_{n,i} : n \in \mathbb{N}, 1 \leq i \leq m_n\}$. This is countable (countable union of finite sets). Given $x \in X$ and $\varepsilon > 0$, choose n with $1/n < \varepsilon$. Then $x \in B(x_{n,i}, 1/n)$ for some i , so there is $d \in D$ with $d(x, d) < \varepsilon$. Thus D is dense in X and X is separable. $\square$

Example 10.3 ($\mathbb{R} \setminus \mathbb{Q}$ is separable). The set of irrationals $\mathbb{R} \setminus \mathbb{Q}$ is separable.

Proof. Define

$$D := \left\{ q + \frac{\sqrt{2}}{n} : q \in \mathbb{Q}, n \in \mathbb{N} \right\} \subset \mathbb{R} \setminus \mathbb{Q}.$$

It is countable. Let $x \in \mathbb{R} \setminus \mathbb{Q}$ and $\varepsilon > 0$. Choose n with $\sqrt{2}/n < \varepsilon/2$. Choose $q \in \mathbb{Q}$ with $|q - x| < \varepsilon/2$. Then $d = q + \sqrt{2}/n \in D$ and

$$|d - x| \leq |q - x| + \sqrt{2}/n < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Hence D is dense in $\mathbb{R} \setminus \mathbb{Q}$. $\square$

11 Lecture 10: Banach Fixed Point Theorem

Definition 11.1 (Contraction mapping). A function $T : (X, d) \rightarrow (X, d)$ is a *contraction* if there exists $c \in (0, 1)$ such that

$$d(Tx, Ty) \leq c d(x, y) \quad \forall x, y \in X.$$

Theorem 11.2 (Banach contraction principle (complete proof)). *If (X, d) is complete and $T : X \rightarrow X$ is a contraction, then:*

- (a) T has a unique fixed point $x^* \in X$ (i.e. $Tx^* = x^*$).
- (b) For any $x_0 \in X$, the iteration $x_{n+1} = Tx_n$ converges to x^* .

Proof. Fix $x_0 \in X$ and define $x_{n+1} = Tx_n$.

Step 1: estimate successive differences. By contraction,

$$d(x_{n+1}, x_n) = d(Tx_n, Tx_{n-1}) \leq c d(x_n, x_{n-1}).$$

Iterating,

$$d(x_{n+1}, x_n) \leq c^n d(x_1, x_0).$$

Step 2: show (x_n) is Cauchy. For $m > n$,

$$\begin{aligned} d(x_m, x_n) &\leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \\ &\leq \sum_{k=n}^{m-1} c^k d(x_1, x_0) \leq d(x_1, x_0) \sum_{k=n}^{\infty} c^k = d(x_1, x_0) \frac{c^n}{1-c}. \end{aligned}$$

As $n \rightarrow \infty$, the right-hand side $\rightarrow 0$, so (x_n) is Cauchy.

Step 3: existence of fixed point. Since X is complete, there exists $x^* \in X$ with $x_n \rightarrow x^*$. We show $Tx^* = x^*$. Note that T is Lipschitz, hence continuous:

$$d(Tx, Ty) \leq c d(x, y) \Rightarrow x_n \rightarrow x \implies Tx_n \rightarrow Tx.$$

Thus

$$Tx^* = T(\lim x_n) = \lim Tx_n = \lim x_{n+1} = x^*.$$

Step 4: uniqueness. If $Tx = x$ and $Ty = y$, then

$$d(x, y) = d(Tx, Ty) \leq c d(x, y).$$

Since $0 < c < 1$, we get $(1-c)d(x, y) \leq 0$, so $d(x, y) = 0$ and $x = y$.

Step 5: convergence to the fixed point. We already proved $x_n \rightarrow x^*$, so the iteration converges to the unique fixed point. $\square$

Remark 11.3 (Why completeness matters). If X is not complete, a contraction may fail to have a fixed point in X because the Cauchy iteration may converge in a completion but not inside X .

12 Lecture 11: Cantor Intersection and Perfect Sets (optional enrichment)

Definition 12.1 (Diameter). For $A \subset X$, define the *diameter*

$$\text{diam}(A) := \sup\{d(x, y) : x, y \in A\} \in [0, \infty].$$

Theorem 12.2 (Cantor intersection theorem). Let (X, d) be complete and let $F_1 \supset F_2 \supset \dots$ be nonempty closed sets with $\text{diam}(F_n) \rightarrow 0$. Then $\bigcap_{n=1}^{\infty} F_n$ consists of exactly one point.

Proof. Pick $x_n \in F_n$. For $m > n$, we have $x_m, x_n \in F_n$, so

$$d(x_m, x_n) \leq \text{diam}(F_n) \xrightarrow{n \rightarrow \infty} 0.$$

Thus (x_n) is Cauchy. Completeness gives $x_n \rightarrow x \in X$.

Fix n . Then for all $m \geq n$, $x_m \in F_n$. Since F_n is closed, it contains the limit x . Hence $x \in \bigcap_n F_n$, so the intersection is nonempty.

If $y \in \bigcap_n F_n$, then $x, y \in F_n$ for every n , so $d(x, y) \leq \text{diam}(F_n) \rightarrow 0$, hence $x = y$. $\square$

Definition 12.3 (Perfect set). A set $P \subset X$ is *perfect* if it is closed and every point of P is a limit point of P .

13 Appendix: A Quick Checklist of Core Results

Remark 13.1. For a complete “Metric Space Notes” file, the following are core pillars (all proved above):

- Metric axioms; reverse triangle inequality; two-point inequality.
- Open sets, closed sets; balls; closure, interior, boundary; algebraic laws.
- Sequences: convergence, uniqueness of limit, Cauchy, convergent $\Rightarrow$ Cauchy.
- Completeness: definition; $\mathbb{Q}$ incomplete; ℓ_p complete; closed subspace $\iff$ complete (inside complete space).
- Continuity: ε - δ , sequential criterion, open-preimage criterion.
- Uniform continuity: definition; Lipschitz; preserves Cauchy; Heine–Cantor on compact sets.
- Compactness: open-cover definition; compact $\Rightarrow$ totally bounded; compact $\Rightarrow$ sequentially compact; compact $\Rightarrow$ closed and bounded.
- Sequential compactness $\Leftrightarrow$ completeness + total boundedness.
- Total bounded $\Rightarrow$ separable.
- Banach contraction principle.